NOT MEASUREMENT SENSITIVE

MIL-PRF-81352B
31 JULY 1997
SUPERSEDING
MIL-L-81352A(AS)
10 February 1972

PERFORMANCE SPECIFICATION

COATING, AIRCRAFT TOUCH-UP

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains the requirements for a general touch-up coating for use on items previously finished in accordance with MIL-F-7179. The volatile organic compound (VOC) content of this coating is less than 340 grams/liter (g/l).

1.2 Classification.

1.2.1 Type. The coating will be of the following types:

Type I - Acrylic base

Type II - Alkyd base

Type III - Polyurethane base

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Air Warfare Center Aircraft Division Code 414100 B120-3, Highway 547, Lakehurst, NJ 08733-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8010

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-P-1892	-	Paint, Varnish, Lacquer, and Related Materials: Packaging, Packing, and Marking of
QQ-A-250/5	-	Aluminum Alloy Alclad 2024, Plate

DEPARTMENT OF DEFENSE

MIL-H-5606	-	Hydraulic Fluid, Petroleum Base, Aircraft, Missile and Ordnance
MIL-PRF-5624	-	Turbine Fuel, Aviation, Grades JP-4, JP-5, and JP-5/JP-8 ST
MIL-F-7179	-	Finishes, Coatings, and Sealants for the Protection of Aerospace Weapons System
MIL-A-8625	-	Anodic Coatings, For Aluminum and Aluminum Alloys
MIL-PRF-22750	-	Coating, Epoxy, High-Solids
MIL-PRF-23377	-	Primer Coatings: Epoxy, High-Solids
MIL-PRF-23699	-	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base, NATO Code Number 0-156
MIL-C-81706	-	Chemical Conversion Materials for Coating Aluminum and Aluminum Alloys
MIL-T-81772	-	Thinner, Aircraft Coatings
MIL-H-83282	-	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Aircraft, Metric, NATO Code Number H-537
MIL-P-85582	-	Primer Coatings: Epoxy, Waterborne

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STANDARDS

FEDERAL

FED-STD-141	-	Paint, Varnish, Lacquer and Related Materials: Methods of Inspection, Sampling and Testing
FED-STD-313	-	Material Safety Data, Transportation Data and Disposal Data For Hazardous Materials Furnished To Government Activities
FED-STD-595	-	Colors Used in Government Procurement

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

FEDERAL REGULATIONS

29 CFR 1910.1200	-	Occupational Safety and Health Standards - Hazard Communications

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are (DoD) adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI-Z129.1	-	Precautionary Labeling of Hazardous Industrial Chemicals, American National Standard for the (DoD Adopted)

(Application for copies should be addressed to American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR QUALITY CONTROL (ASQC)

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ASQC-Z1.4 - Sampling Procedures and Tables for Inspection by Attributes (DoD Adopted)

(Application for copies should be addressed to the American Society for Quality Control, P.O. Box 3005, 611 East Wisconsin Avenue, Milwaukee, WI 53201-4606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D185 - Coarse Particles in Pigments, Pastes, and Paints, Standard Test Method for (DoD Adopted)
ASTM-D522 - Mandrel Bend Test of Attached Organic Coatings, Standard Test Method for (DoD Adopted)
ASTM-D523 - Specular Gloss, Standard Test Method for (DoD Adopted)
ASTM-D823 - Producing Films of Uniform Thickness of Paint, Varnish, and Related Products on Test Panels, Standard Test Methods for (DoD Adopted)
ASTM-D1200 - Viscosity by Ford Viscosity Cup, Standard Test Method for (DoD Adopted)
ASTM-D1210 - Fineness of Dispersion of Pigment-Vehicle Systems, Standard Test Method for (DoD Adopted)
ASTM-D1296 - Odor of Volatile Solvents and Diluents, Standard Test Method for (DoD Adopted)
ASTM-D1308 - Effect of Household Chemicals on Clear and Pigmented Organic Finishes, Standard Test Method for (DoD Adopted)
ASTM-D1729 - Visual Evaluation of Color Differences of Opaque Materials, Standard Practice for (DoD Adopted)
ASTM-D2244 - Calculation of Color Differences from Instrumentally Measured Color Coordinates, Standard Test Method for (DoD Adopted)
ASTM-D2805 - Hiding Power of Paints by Reflectometry, Standard Test Method for (DoD Adopted)
ASTM-D3335 - Low Concentrations of Lead, Cadmium, and Cobalt in Paint by Atomic Absorption Spectroscopy, Standard Test Method for (DoD Adopted)
ASTM-D3718 - Low Concentrations of Chromium in Paint by Atomic Absorption Spectroscopy, Standard Test Method for (DoD Adopted)
ASTM-D3924 - Standard Environment for Conditioning and Testing of Paint, Varnish, Lacquer and Related Materials, Standard Specification for (DoD Adopted)
ASTM-D3960 - Volatile Organic Compound (VOC) Content of Paints and Related Coatings, Standard Test Method for (DoD Adopted)
ASTM-E275 - Describing and Measuring Performance of Ultraviolet, Visible, and Near-Infrared Spectrophotometers, Standard Practice for

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ASTM-G26 - Operating Light-Exposure Apparatus (Xenon-Arc Type) With and Without Water for Exposure of Nonmetallic Materials, Standard Practice for (DoD Adopted)

(Application for copies should be addressed to American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Materials used to manufacture these coatings are not limited. The resulting coating(s) shall conform to the requirements of this specification.

3.2 Toxicity. The coating shall have no adverse effect on the health of personnel when used for their intended purpose (see 6.1). A Material Safety Data Sheet (MSDS) shall be prepared in accordance with FED-STD-313 and 29 CFR 1910.1200 (see 6.3).

3.3 Composition. The coating shall be a one component composition and shall not contain chromium, when tested in accordance with ASTM-D3718, nor cadmium, cadmium compounds, or more than 0.06 percent by weight of lead metal or lead compounds, when tested in accordance with ASTM-D3335 (see 4.5).

3.3.1 Volatile organic compound (VOC) content. When mixed in accordance with the manufacturer's instructions to achieve the application viscosity specified in 3.5.4, the VOC content of the coating shall be not greater than 340 g/l, when tested in accordance with ASTM-D3960 (see 4.5). Halogenated solvents are prohibited.

3.3.2 Thinner compatibility. The coating shall be miscible in water or compatible with the following solvents: type I and II coatings shall be compatible with MIL-T-81772, type III; and type III coating shall be compatible with MIL-T-81772, type I.

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3.4 Storage stability.

3.4.1 Extended storage stability. When tested in accordance with FED-STD-141, method 3022, "Storage Stability (Filled Containers)," the coating, as packaged by the manufacturer, shall be capable of meeting all requirements specified herein for a minimum of one year from the date of manufacture when stored under the following conditions (see 4.5):

a. The daily mean ambient temperature at the storage point is 35 to 95° F (1.7 to 35° C)
b. The peak ambient air temperature at the storage point does not exceed 115° F (46° C)

3.4.2 Accelerated storage stability. The coating, as packaged by the manufacturer, shall not develop increased internal container pressure when tested in accordance with 4.5.1,. After storage, the contained material shall be free of lumps, skins, gels, or particulate matter, either suspended in solution or settled on the inner surface of the container. When stirred with a paddle, the contained material shall be a smooth, homogenous mixture. When applied in accordance with 4.4.2, the coating shall be a smooth, uniform film, free of grains, lumps and streaks.

3.5 Physical properties.

3.5.1 Condition in container. Upon opening a full, previously unopened container, the coating shall exhibit no grit, skinning, curdling, livering or excessive pigment flotation, and shall exhibit no settling or caking that may be dispersed by mixing with a hand paddle to a uniform homogeneous condition, in accordance with FED-STD-141, method 3011, "Condition in Container" (see 4.5).

3.5.2 Coarse particle content. The particles retained on a No. 325 sieve shall not be greater than 0.1 percent by total weight of the coating, when tested in accordance with ASTM-D185 (see 4.5).

3.5.3 Fineness of grind. The fineness of grind of the coating shall be not less than 7.0 for gloss colors and not less than 5.0 for all other colors when tested in accordance with ASTM-D1210 (see 4.5).

3.5.4 Viscosity. The viscosity of the coating, as prepared for application in accordance with the manufacturer's instructions, shall conform to table I when tested in accordance with ASTM-D1200 (see 4.5).

TABLE I. Viscosity requirements.

Color	Time through a number 4 Ford cup
Clear	14 to 25seconds
All other colors	16 to 25seconds

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3.6 Working properties. The coating shall yield a smooth and uniform film when applied by spray, brush, or roller and shall conform to 3.6.1 through 3.6.4.

3.6.1 Odor. The odor of the coating, as a wet or dry film, shall not be obnoxious when tested in accordance with ASTM-D1296 (see 4.5). The air-dried coating shall retain no residual odor 48 hours after application.

3.6.2 Drying time. The coating, when applied to a wet film thickness of 1.6 to 2.4 mil [40 to 60 microns (µm)], shall dry hard within 40 minutes, when tested in accordance with FED-STD-141, Method 4061, "Drying Time" (see 4.5).

3.6.3 Surface appearance. The coating shall dry to a smooth, uniform film, free of runs, sags, bubbling, wrinkling, streaking, blushing, haze, blisters, coarse particles, silking, or other surface irregularity when prepared and examined in accordance to 4.5.2 (see 4.5).

3.6.4 Coating system compatibility. All coatings supplied under this specification shall be compatible with the coatings listed in MIL-F-7179, table I.

3.7 Film properties.

3.7.1 Color. When tested in accordance with ASTM-D1729 (see 4.5), the color of the coating shall be a visual match with the specified FED-STD-595 color chip (see 6.2 and 6.4.3).

3.7.2 Specular gloss. The applied coating shall conform to table II when tested at a 60° geometry in accordance with ASTM-D523 (see 4.5).

TABLE II. Specular gloss.

	Minimum	Maximum
Gloss	80	---
Semi-gloss	15	45
Camouflage	---	5

3.7.3 Infrared reflectance (applicable only to FED-STD-595, color number 34095). The total infrared reflectance, specular and diffuse, of the coating shall be not greater than 8, when tested in accordance with 4.5.4.

3.7.4 Adhesion (wet). The coating, applied to test panels in accordance with 4.4 through 4.4.2, and then immersed in distilled water for 24 hours, shall not peel away from the test panel when tested in accordance with FED-STD-141, Method 6301, "Adhesion (Wet) Tape Test," (see 4.5 and table IV).

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3.7.5 Opacity. The coating, cast to a wet film thickness of 1.7 to 2.3 mils (43 to 58 µm) on a black and white chart and tested in accordance with ASTM-D2805 (see 4.5), shall have a contrast ratio of not less than 0.95 for all colors, with the exception of FED-STD-595, color number, which shall have a contrast ratio of not less than 0.90.

3.7.6 Flexibility. The coating, applied to test panels in accordance with 4.4 through 4.4.2, shall exhibit no cracking, peeling, or loss of adhesion, when bent over a 1.0 in. (25.4 mm) mandrel, in accordance with ASTM-D522, method B, "Cylindrical Mandrel Test" (see 4.5 and table I).

3.8 Resistance properties.

3.8.1 Fluid resistance. The coating, applied to test panels in accordance with 4.4 through 4.4.2, and completely immersed in fluids in accordance with table III, shall exhibit no blistering, softening, or other coating defects four hours after removal from the fluids. A slight staining of the coating is acceptable.

TABLE III. Fluid immersion conditions.

Fluid	Fluid temperature	Time of immersion	
Lubricating oil conforming to MIL-PRF-23699	250 ±5°F (121 ±3°C)	Type I Type II Type III	4 hours 4 hours 24 hours
Hydraulic fluid conforming to MIL-H-5606	150 ±3°F (66 ±3°C)	24 hours	
Hydraulic fluid conforming to MIL-H-83282	150 ±3°F (66 ±3°C)	24 hours	
JP-5 fuel conforming to MIL-PRF-5624	Room temperature	7 days	

3.8.2 Weather resistance. The coating shall have a color difference value [delta E (ΔE)] of not greater than 2.0, as compared to the unexposed coating, when tested in accordance with 4.5.5.

3.8.3 Heat resistance. The coating shall not exhibit a color change (ΔE shall be not greater than 1.0) as compared to the unexposed control coating, when tested in accordance with 4.5.6.

3.8.4 Water resistance. The coating shall exhibit no cracking, blistering, or whitening when tested in accordance with 4.5.7. A slight whitening or dulling, which may be removed by lightly wiping with a soft cloth, shall not be cause for rejection.

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3.9 Marking and labeling. All containers that serve as shipping containers shall be marked in accordance with ANSI-Z129.1. Individual containers shall be marked in accordance with PPP-P-1892 and shall bear permanent labels showing the following:

- Specification MIL-PRF-81352B, "Coating, Aircraft Touch-up"
- Type number and name (see 1.2.1)
- Color (FED-STD-595 color name and number)
- Manufacturer's name, product identification, and lot number
- Date of manufacture (month/year)
- VOC content (g/l) and type of thinner compatible with the coating
- Net contents
- Mixing instructions

4. VERIFICATION

4.1 Classification of inspection. The inspection requirement specified herein is classified as:
a. Conformance inspection (see 4.2).

4.2 Conformance inspection. The conformance inspection shall consist of all examinations and tests specified in table IV. On a previously tested product, certification of conformance to 3.4.1 may be accepted in place of conducting the test; however, the supplied material shall be tested for conformance to all other requirements. In the event that a test sample fails to meet any requirement of this specification, the lot (see 4.2.1) represented by the sample shall be rejected.

4.2.1 Lot formation. A lot shall consist of all the same type and color of coating, manufactured at one time, forming part of one contract or order, and submitted for acceptance to the procuring activity.

4.2.2 Report of tests. When required (see 6.2), the manufacturer shall submit to the procuring activity a certified test report for each lot of material. The test report shall show the results of the conformance tests of this specification.

4.3 Sampling.

4.3.1 Sampling for tests. Samples shall be selected in accordance with FED-STD-141, Method 1022, "Sampling for Inspection and Testing."

4.3.2 Sampling of filled containers. Samples shall be selected at random from each lot in accordance with ASQC-Z1.4.

4.4. Test panels. Test panels shall be prepared under laboratory conditions (see 4.5). The approximate dimensions for all test panels shall be 0.020 x 3 x 6 in. (0.5 x 76.2 x 152.4 mm).

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Unless otherwise specified in the test method, the test panels shall be Alclad aluminum alloy 2024 (T3 temper) conforming to QQ-A-250/5.

4.4.1 Panel preparation. The panels shall be treated with conversion coating conforming to MIL-C-81706, class 1A.

4.4.2 Application of coatings.

a. Primer coating. One cross-coat of a primer coating conforming to MIL-PRF-23377 or MIL-P-85582 shall be spray-applied to a dry film thickness of 0.6 to 0.9 mils (15 to 23 µm) in accordance with ASTM-D823. Air-dry MIL-P-85582 for a minimum of one hour prior to applying additional coatings. Air-dry MIL-PRF-23377 for a minimum of four hours prior to applying additional coatings.

b. Topcoat. Prepare the coating in accordance with the manufacturer's instructions (the admixed topcoat may be reduced with thinner conforming to MIL-T-81772 if so instructed by the manufacturer). If reducing with thinner, do not exceed the maximum allowable VOC content (see 3.3.1). Spray apply the coating in accordance with ASTM-D823 to a dry film thickness of 1.7 to 2.3 mils (43 to 58 µm). Prior to testing, the applied coating shall be air-dried for a minimum of 14 days at room temperature.

4.5 Test methods. The tests of this specification shall be conducted in accordance with table IV and 4.5.1 through 4.5.7 with test panels prepared in accordance with 4.4 through 4.4.2, when directed. Unless otherwise specified in the test method or paragraph, all testing shall be conducted under the laboratory testing conditions in accordance with ASTM-D3924 and in duplicate. Unless otherwise stated in the test method or paragraph, room temperature shall be defined as 70 ±10°F (21 ±5°C) and relative humidity of 50 ±10 percent.

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TABLE IV. Requirements and test methods.

Requirement Paragraph	Requirement	Test paragraph	FED-STD-141 Method Number	ASTM Method
3.1	Material	---	---	---
3.2	Toxicity	---	---	---
3.3	Composition			
	Cadmium and lead content	---	---	D3335
	Chromium content	---	---	D3718
3.3.1	VOC Content			D3960
3.3.2	Thinner Compatibility	---	---	---
3.4.1	Extended Storage Stability	---	3022	---
3.4.2	Accelerated Storage Stability	4.5.1	---	---
3.5.1	Condition in Container	---	3011	---
3.5.2	Coarse Particle Content	---	---	D185
3.5.3	Fineness of Grind	---	---	D1210
3.5.4	Viscosity	---	---	D1200
3.6	Working Properties	---	---	---
3.6.1	Odor	---	---	D1296
3.6.2	Drying Time	---	4061	---
3.6.3	Surface Appearance	4.5.2	---	---
3.6.4	Coating system compatibility	---	---	---
3.7.1	Color	---	---	D1729
3.7.2	Specular Gloss	4.5.3	---	D523
3.7.3	Infrared Reflectance	4.5.4	---	---
3.7.4	Adhesion	---	6301	---
3.7.5	Opacity	---	---	D2805
3.7.8	Flexibility	---		D522
3.8.1	Fluid Resistance	Table III	---	---
3.8.2	Weather Resistance	4.5.5	---	G26
3.8.3	Heat Resistance			
	Type I	4.5.6	6051 & 6304	D2244
	Type II & III	4.5.6	6051	D2244
3.8.4	Water Resistance	4.5.7	---	D1308
3.9	Marking and labeling	---	---	---

4.5.1 Accelerated storage stability. The coating, in its original, unopened container, shall be stored at 125 ±3°F (52 ±2°C) for a minimum of 14 consecutive days. During the storage period, the container should be placed in a larger, vented container to confine any splashing that may occur if the lid of the unopened container is blown off by increased internal pressure. At the end of the 14 day period, remove the container and allow it to cool to room temperature. If the unopened container is deformed upon removal from storage, do not open and discard. If it is not deformed, open the container carefully and examine its contents for conformance to 3.4.2.

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4.5.2 Surface appearance. The coating shall be applied to test panels in accordance with 4.4 through 4.4.2 and examined under a magnification of 10 to 15 diameters for conformance to 3.6.3.

4.5.3 Specular gloss. Test panels shall conform to QQ-A-250/5, with approximate dimensions of 0.020 x 3.0 x 6.0 in. (0.5 x 76.2 x 152.4 mm), and shall be anodized in accordance with MIL-A-8625, type I. The coating shall be applied directly to the unprimed test panels and air-dried for 14 days. Examine the in accordance with ASTM-D523 at a 60° geometry for conformance to 3.7.2.

4.5.4 Infrared reflectance (applicable only to FED-STD-595, color number 34095). The coating, whose color conforms to FED-STD-595, color number 34095, shall be applied to test panels in accordance with 4.4 through 4.4.2. The total infrared reflectance of the coating, specular and diffuse, shall be tested relative to barium sulfate in the wavelength range of 700 to 2600 nanometers (nm) with a spectrophotometer (Perkin-Elmer LAMBDA 9, or equivalent). ASTM-E275 may be used to compare performance of the equivalent unit to the LAMBDA 9. Examine for conformance to 3.7.3.

4.5.5 Weather resistance. Prepare three test panels in accordance with 4.4 through 4.4.2. One test panel shall be used as the control. The other two test panels shall be exposed for 500 hours in a Xenon-arc weatherometer (Atlas Electric Devices Company or equivalent) that is cycling between 102 minutes of light only and 18 minutes of light and water spray. After exposure, measure the color difference (ΔE) value of the exposed panels and the control and examine the coating for conformance to 3.8.2. The following conditions shall apply when testing in accordance with ASTM-G26, type BH:

a. Black body temperature in cabinet: 140 ±5°F (60 ±3°C).
b. Relative humidity in cabinet: 50 ±5 percent.
c. Intensity (spectral irradiance) of Xenon-arc: 0.35 ±0.01 watts/meter^2 (W/m^2) at a wavelength of 340 nm.

4.5.6 Heat resistance. Prepare two panels in accordance with 4.4 through 4.4.2. One test panel shall act as the control, the other as the test specimen. Expose the test specimen for four hours at 250 ±5°F (121 ±3°C) in accordance with FED-STD-141, method 6051, "Heat Resistance." After exposure, examine the test specimen and the unexposed control in accordance with ASTM-D2244 for conformance to 3.8.3.

4.5.7 Water resistance. Test panels shall be prepared in accordance with 4.4 through 4.4.2, with the exception the applied coatings shall be air-dried for seven days at room temperature. The test panels shall then be immersed in distilled water in accordance with ASTM-D1308 for 24 hours at room temperature. Within five minutes of removal from the water, the coating shall be examined for conformance to 3.8.4.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful but is not mandatory.)

6.1 Intended use. The coating conforming to this specification is intended to be an exterior coating for touch-up purposes over epoxy primer conforming to MIL-PRF-23377 or MIL-P-85582. All types (see 1.2.1) of this coating may be used as an insignia and marking coating directly over freshly applied epoxy topcoat conforming to MIL-PRF-22750. The type I (acrylic) coating may be applied over polyurethane coating conforming to MIL-PRF-85285; however, a thin cross-coat of epoxy primer is necessary over the polyurethane prior to its application. Scuff-sand any aged epoxy coating before applying this coating.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification, including any amendments.
b. Issue of DoDISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.2 and 2.3).
c. Type and color (see 6.4).
d. Quantity, size and type of containers in which the coating is to be furnished (see 6.4).
e. Whether certification of conformance or the performance of the extended storage stability test is required (see 4.2).
f. Test or inspection report, if required (see 4.2.2).

6.3 Material Safety Data Sheet (MSDS). 29 CFR 1910.1200 requires that the MSDS for each hazardous chemical used in an operation must be readily available to personnel using the material. Contracting officers should identify the activities requiring copies of the MSDS.

6.4 Part or identify number (PIN). Part numbers may be coded in accordance with table VI.

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TABLE VI. PIN definition (see 6.4).

M81352	-	X	-	XX	-	XXXXX
Specification identification		Type designator Type I = 1 Type II = 2 Type III = 3		Container size designator 1 pint (1.88 l) = 1P 1 quart (3.76 l) = 1Q 1 gallon (75.7 l) = 1G		Color designation (see table VII)

TABLE VII. Selected FED-STD-595 color numbers and common names.

Gloss Colors		Camouflage Colors	
Color Number	Color Name	Color Number	Color Name
--	Clear	26081	Seaplane Gray
10049	Maroon	27875	Semi-gloss Insignia White
11136	Insignia Red	31136	Insignia Red
12197	International Orange	33538	Orange Yellow
13538	Orange Yellow	34087	Olive Drab
13655	Light Yellow	34097	Field Green
14087	Olive Drab	34151	Interior Green
14187	Light Green	35042	Non-specular Sea Blue
15044	Insignia Blue	35044	Insignia Blue
15102	Light Blue	35237	Blue Gray
16081	Engine Gray	36231	Dark Gull Gray
16440	Light Gray	36320	Dark Gray
16473	Aircraft Gray	36375	Medium Gray
17038	Gloss Black	36440	Light Gull Gray
17875	Insignia White	37038	Black
17925	Untinted White	37875	Insignia White

6.5 Subject term (Keyword) listing.

Acrylic coating
Alkyd coating
Polyurethane coating

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - MR
Navy - AS
Air Force - 11
GSA - FSS

Review:
Army - MD1
Navy -CG
Air Force - 70

Preparing activity:
Navy - AS
(Project 8010-0001)

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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-81352B	2. DOCUMENT DATE (*YYMMDD*) 970731

3. DOCUMENT TITLE

COATING, AIRCRAFT, TOUCH-UP

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial: (2) DSN (*If Applicable*):	7. DATE SUBMITTED (*YYMMDD*)

8. PREPARING ACTIVITY

a. NAME COMMANDER NAVAL AIR WARFARE CENTER AIRCRAFT DIVISION	b. TELEPHONE NUMBER (*Include Area Code*) (1) Commercial: (908) 323-7488 (2) DSN: 624-7488
c. ADDRESS (*Include Zip Code*) CODE 414100B120-3 HIGHWAY 547 LAKEHURST, NJ 08733-5100	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone: (703) 756-2340 DSN 289-2340

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NAVAL AIR WARFARE CENTER
AIRCRAFT DIVISION
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LAKEHURST, NJ 08733-5100

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